*Research paper*

**Chapter 16**

# Some Eco-physiological Characters of *Austroeupatorium inulifolium* (Kunth) R.M. King & H. Rob.

Inoka Piyasinghe[1], Sumedha M. Weerasinghe[2] and Jagath Gunatilake[2]
[1]Postgraduate Institute of Science, University of Peradeniya, Sri Lanka
[2]Faculty of Science, University of Peradeniya, Sri Lanka

**Abstract:** Alien plants are invading the natural and semi-natural ecosystems causing much damage to the biodiversity. For the past few years, *Austroeupatorium inulifolium* has invaded aggressively within the Knuckles Conservation Area of Sri Lanka. However, *A. inulifolium* is still to be identified as an invasive plant in Sri Lanka, hence no scientifically valid data are available on their impact on the ecosystem. The baseline information on the ecology of *A. inulifolium* is important to undertake future management practices successfully. In the present experiment the morphological plasticity of *A. inulifolium* when exposed to different nutrient pulses and intra-specific competition was investigated. Two pot experiments were set up under glasshouse conditions at the Department of Botany, University of Peradeniya, Sri Lanka. In the first experiment, seedlings were transplanted (1 plant/pot) and nutrients were added to create 'spatial', 'temporal' and 'homogeneous' pulses for five weeks. After 8 weeks of growth, plants were harvested. In the second experiment, seedlings were exposed to 3 density levels; 1, 2, and 4 seedlings/pot (x1, x2 and x4 respectively; n=10). Plants were harvested after 9 weeks. Results were statistically analyzed using Minitab 14. *Austroeupatorium inulifolium* seedlings exposed to 'temporal' nutrient enrichment showed higher shoot, root, total biomass and growth rates than plants given with homogenous nutrient-enrichment, suggesting plants have the ability to capture unpredictable 'temporal' pulses more efficiently than predictable nutrient pulses. However, the root weight ratio (RWR) and root:shoot ratio showed no significant difference between treatments indicating that their better growth under temporal pulses is mainly through physiological adjustments rather than morphological. In the second experiment, shoot, root and total biomass at high density (x4) showed significantly lower values ($p<0.05$) than the low density treatments. The relative height ratio and growth rate of *A. inulifolium* also showed similar trends to biomass measurements. *A. inulifolium* responded to intra-specific competition by lowering its growth with increase in competition. The results suggest that *A. inulifolium* is physiologically more plastic to

---

Inoka Piyasinghe, Sumedha M. Weerasinghe and Jagath Gunatilake (2010): Some Eco-physiological characters of *Austroeupatorium inulifolium* (Kunth) R.M. King & H. Rob. In: Invasive Alien Species in Sri Lanka – Strengthening Capacity to Control Their Introduction and Spread (Eds: Marambe, B., Silva, P., Wijesundara, S. and Atapattu, N.), pp 151-160. Biodiversity Secretariat of the Ministry of Environment, Sri Lanka.

unpredictable nutrient pulses than predictable pulses (spatial and homogeneous), which is considered as an important competitive trait possess by invasive species.

## Introduction

Invasive species persist and spread in ecosystems and alter the basic ecosystem processes such as hydrology, nutrient cycling, soil erosion rates (Vitousek and Walker, 1989), fire frequency and intensity (D'Antonio and Vitousek, 1992), and recruitment of native species (Mooney and Drake, 1989). A number of introduced non-native (alien) species also can become locally dominant (Levine *et al*., 2003) and change relatively diverse communities into near monocultures. Natural and semi-natural ecosystems of Sri Lanka have been threatened by the invasion of alien species and their rapid distribution is mostly attributed to human activities. Invasive alien species (IAS) are able to survive under a wide range of climatic and soil conditions, and reproduce, spread at alarming rates across the landscape. Eco-physiological information of alien invasive plants and their impact on native plants are scarce. Scientific information on these aspects would be imperative to assess the damages and develop strategies to overcome the problems.

Experimental and inadvertent manipulations of resources have shown that the availability of resources such as mineral nutrients and water are important determinants of the invasibility of communities by plants. Adding nutrients (Wedin and Tilman, 1996; Bakker and Berendse, 1999) and water often promotes invasion (Milchunas and Lauenroth, 1995; White *et al*., 1997). Decreasing soil water availability has also promoted invasion (Cleverly *et al.,* 1997).

Soil nutrient patches are spatial or temporal in nature (Fitter, 1994). Plants respond to such environmental heterogeneity by altering growth and/or adjusting biomass partitioning to various organs (Dale and Causton, 1992; Reynolds and D'Antonio, 1996). Plants seem to adjust physiologically to capture unpredictable, short-term temporal pulses, while long-duration, predictable nutrient pulses are captured through morphological adjustments. Fransen *et al*. (2001) suggest that physiological plasticity is more critical for long-term competitive advantage of a plant than morphological plasticity.

According to Casper *et al*. (2000), localization of nutrients within patches would increase the intensity of competition, as nutrient-rich patches can contain roots of several plants. Nitrogen capture by competing plants in nutritionally heterogeneous conditions strongly correlates with their rates of root proliferation in nitrogen-rich patches. Thus, the species with faster root proliferation gain a competitive advantage over the other plants (Robinson *et al*., 1999). This also suggests that nutrient heterogeneity may affect intensity of competition.

The competitive ability of invasive plants could be different when they are in a mixed population rather than in monoculture stands. When plants face competition, they tend to adjust morphologically and physiologically. Studies suggest that invasive species may be more plastic than non-invasive species as this will decide their competitive advantage over the neighbors (Williams *et al*., 1995; McDowell, 2002). Populations in the introduced range of an invasive species have evolved greater plasticity than populations in the native range because of their high competitive ability over other species (Sexton *et al*., 2002; Parker *et al.,* 2003).

*Austroeupatorium inulifolium* was introduced to Asian countries during the World War II, and Sri Lanka and the Western Ghats in India were infested by this plant probably through military troops and their equipment returning from Assam and the Burma front (McFadyen, 2003). The

species has aggressively invaded the Knuckles Conservation Area of Sri Lanka in the recent years invading many landuse types including forest clearings, disturbed lands, grasslands, plantations, forest edges and roadsides forming dense thickets and choking out other vegetation. No vegetation survives beneath the *A. inulifolium* plants and tree seedlings cannot penetrate the dense growth. Grasslands have also been invaded, resulting in complete shading out of the grasses. Following fire hazards (during the dry season), the *A. inulifolium* plants show rapid establishment through induced seed germination. Knowledge on the eco-physiological features of *A. inulifolium* could be used to adopt judicious management strategies to control its invasion.

This paper reports the extent of morphological and/or physiological plasticity of *A. inulifolium* when exposed to heterogeneous nutrient pulses and intra-specific competition.

## Materials and Methods

*Austroeupatorium inulifolium* (Kunth) R.M. King & H. Rob. (Synonym: *Eupatorium inulifolium* H.B.K.; family – Asteraceae) plants of similar age (3-4 leaved stage) collected from the nursery, were planted in pots (1 plant per pot) filled with a mixture of nutrient rich garden soil and river sand (1:3 ratio). Pots were hypothetically divided into four sections and nutrients were added (ALBERT'S nutrient solution; N 11%, $P_2O_5$ 8%, $K_2O$ 15%, CaO 14% and MgO 4%, Zn, Cu, Mo, Fe, Mn, B are in balanced form) in such a way to create natural nutrient pulses such as 'spatial', 'temporal' and 'homogeneous' in pots (100ml/pot) (Figure 1).

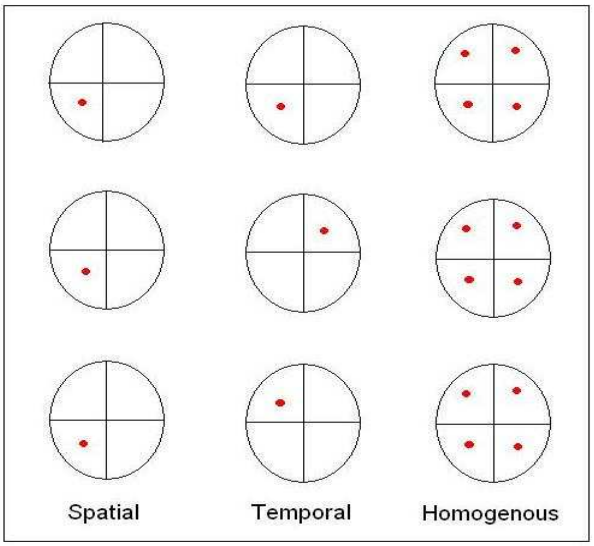


**Figure 1. Nutrient application patterns to create 'spatial', 'temporal' and 'homogeneous' (control) nutrient pulses**

After four weeks of initial growth, nutrient pulse treatments were started and continued once a week for five weeks. To minimize the position effect, pots were randomly rearranged every other week. At the end of 9$^{th}$ week, plants were destructively harvested, and the fresh and dry weights (oven dried at 40°C for 48 hrs) of root and shoot samples were recorded separately.

In the second pot experiment, *Austroeupatorium* seedlings (raised from seeds) were transplanted at three different densities (1, 2, and 4 per pot) in pots containing the mixture of garden soil and sand (1:3 ratio). Plants were placed in the pot equidistant to each other. Each treatment had 10 replicates, totaling to 30 pots. Data were collected from 4 weeks after transplanting. Pots were watered daily and kept at the field capacity. Nine weeks after planting, plants were harvested, fresh weights of root and shoot samples were recorded, and the dry weights were recorded after oven drying at 40°C until a constant weight. The initial seedling weight was calculated using the dry weight of 10 randomly picked seedlings, which were oven dried at 40°C for 48 hrs. The plant heights were recorded every fortnight starting from 4 weeks after planting. The growth rate and root/shoot ratio were also calculated. The root weight ratio (RWR) and relative height ratio (RHR) were calculated using the following equations (Xiao *et al.* 2006).

$$RWR = (\text{root weight}/\text{total weight}) \times 100$$

$$RHR = \frac{[\ln(\text{harvest stem height}) - \ln(\text{initial stem height at transplanting})]}{\text{Time in growth (days)}}$$

**Data analyses**

Data were analyzed using the MS-Excel and the Minitab 14 statistical software packages. Mean values and standard error of means were calculated using MS-Excel software. The results were analyzed using one-way ANOVA and Tukey's family error test was used for mean comparisons.

## Results and Discussion

**Experiment – I**

The shoot and root dry weights (Figure 2) of *A. inulifolium* are significantly higher ($P < 0.05$) in plants exposed to 'temporal' nutrient pulses than plants with homogenous supply of nutrients. Higher root growth indicates root proliferation when plants are given temporal nutrient pulses.

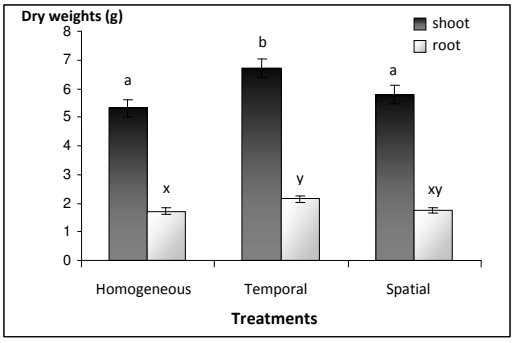


**Figure 2. Root and shoot day weights (g) of *A. inulifolium* in homogenous (control), spatial and temporal nutrient treatments. Different letters indicate significant differences between treatments at p<0.05 levels; for shoot biomass: p=0.019, F=4.86; for root biomass: p=0.028, F=4.29, ANOVA.**

The total biomass (Figure 3) also showed significantly higher values ($p<0.05$) in plants exposed to temporal nutrient-enrichment when compared to plants exposed to homogenous nutrient enrichment.

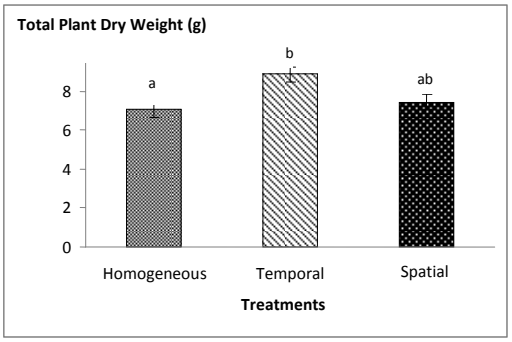


**Figure 3. Total dry weights (g) of *A. inulifolium* in homogenous, spatial and temporal treatments. Significant differences are indicated using different letters (at p=0.05).**

Previous studies have shown that when plants were given heterogeneous nutrient pulses (spatial or temporal), plants acquire and produce more biomass compared to those supplied with homogenous nutrient pulses (Hutchings and de Kroon, 1994). As under field conditions most nutrient pulses are unpredictable (temporal) in nature hence present results suggest that *A. inulifolium* is more effective in acquiring nutrients from unpredictable pulses than predictable pulses (spatial and homogeneous).

The shoot to root (S/R) ratio and RWR showed no significant differences between treatments (Figure 4). This indicates that though plants performed differently under varying nutrient pulses, their biomass allocation pattern to above- and below-ground parts seems to be almost same. This result suggests that *A. inulifolium* did not proliferate roots when in contact with temporal nutrient pulses. The growth rate (Figure 5) also follow similar trend to other parameters tested where plants exposed to temporal patches showed significantly higher rates than that of predictable nutrient pulses such as 'homogeneous' and 'spatial'.

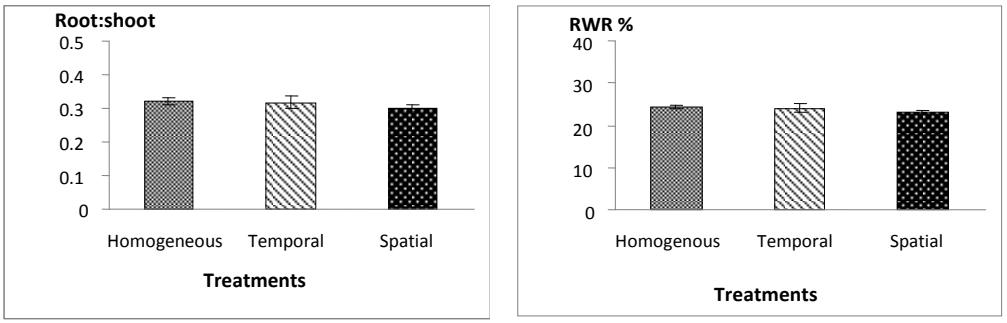


**Figure 4. Shoot:root and root weight ratios of *A. inulifolium* in homogenous, temporal and spatial treatments.**

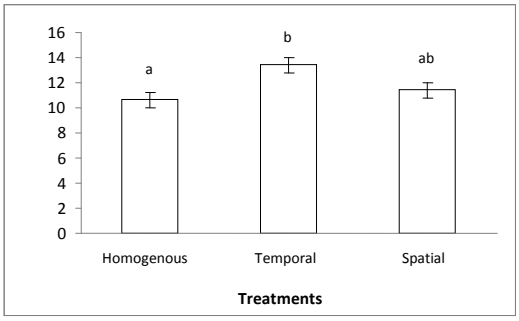


**Figure 5. Growth rate per week *A. inulifolium* in homogenous, spatial and temporal treatments (bars represented by the same letter are not significantly different at p=0.05).**

**Experiment-II**

The performance of *A. inulifolium* when exposed to different density levels (intra-specific competition) are illustrated in Figure 6.

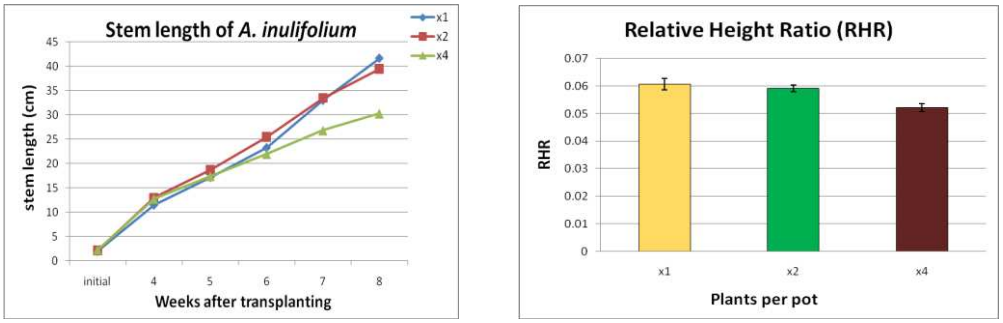


**Figure 6. Stem length (cm) and relative height ratio of *A. inulifolium* under Low (x1), medium (x2) and high density (x4) treatments (bars represented by the same letter are not significantly different at p=0.05).**

At early stage of growth, *A. inulifolium* performed almost similarly and with time (approximately after the 6$^{th}$ week) the plants exposed to high density (x4) showed a decreasing trend in plant height compared to plants exposed to lower density levels (Figure 6). Plants exposed to medium (x2) and low (x1) increased their plant height gradually over time. The relative height ratio (RHR) showed significantly lower values at higher density (x4) compared to lower density levels (x1, x2) (Figure 6). The plants experience limited nutrients with time and plants with high intra-specific competition suffer more compared to low competition. In spite of *A. inulifolium's* aggressive nature they seem to suffer when they are in mono-culture stands due to lack of nutrients and other resources. Under natural environment, large size differences between individuals of large populations are probably due to their intra-specific competition (Meekins and Macarthy, 2000). Xiao *et al.* (2006) suggested that the mean height of plants will decrease with increasing density when the population density reaches a certain value. When there is competition for light between neighbors, height growth will help individual plants avoid the shading effect of its neighbors and enable it to acquire more of the light resources (Schmitt and Wulff 1993; Dudley and Schmitt, 1996; Xiao *et al*., 2006).

The increasing density levels (or intra-specific competition) show decreasing total biomass per plant in *A. inulifolium* (Figure 7). This result also confirms the intensity of intra-specific interactions among individuals which reflects in the plants' overall performance. Grace and Tilman (1990) noted that plant density possibly influence the individual plant biomass due to competition for factors such as nutrient, water, light and physical space. Casper and Cahill (1998) found that overall (i.e. not necessarily individual) biomass of *Abutilon theophrasti* was low when plants were grown in monospecific stands under high density conditions.

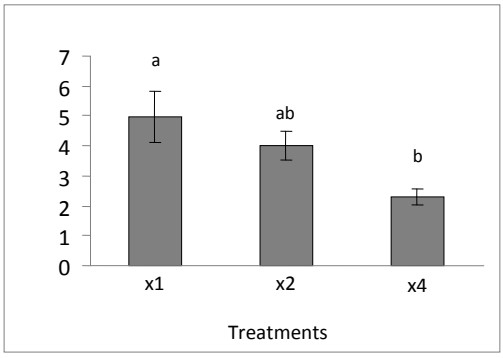


**Figure 7. Total dry biomass (g) of *A. inulifolium* under Low (x1), medium (x2) and high density (x4) treatments after 8 weeks of planting (bars represented by the same letter are not significantly different at p=0.05).**

Shoot and root biomass also depicted similar trends when exposed to different density levels (Figure 8). Weiner (1990) suggested that competitive above-ground advantage is important since above ground interactions are intense as each individual attempts to obtain limiting light resources. There is evidence that shading can reduce root growth and resource uptake within as few as 24 hours (Massimino *et al.,* 1981; Caldwell *et al.,* 1987). However, in the present experiment plants did not show differences in their biomass allocation patterns (root:shoot and RWR) when they exposed to different density levels (Figure 9). Growth rates also showed significantly lower values under high density treatment compared to other treatments (Figure 10).

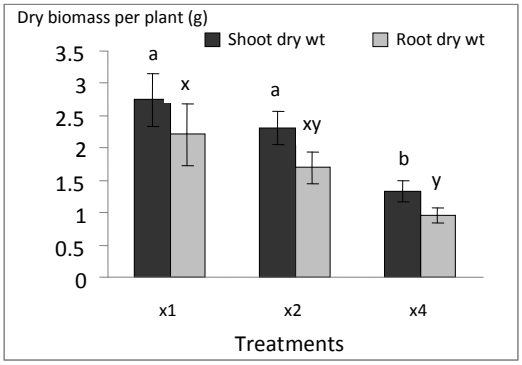


**Figure 8. Root dry weights (g) and shoot day weights of *A. inulifolium* under Low (x1), medium (x2) and high density (x4) treatments (bars represented by the same letter are not significantly different at p=0.05).**

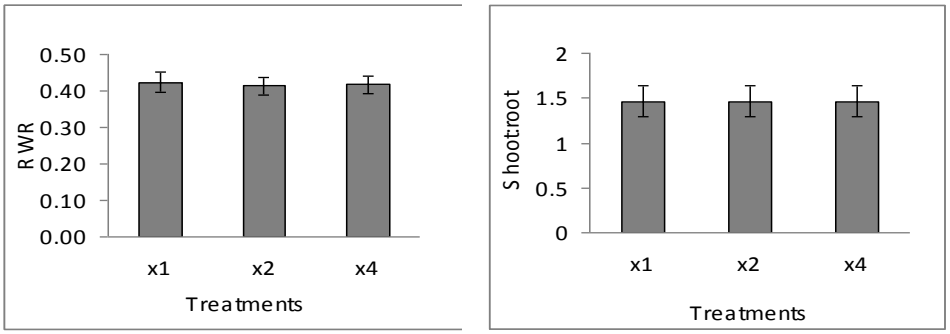


**Figure 9. The RWR and Shoot:root ratio of *A. inulifolium* under Low (x1), medium (x2) and high density (x4) treatments.**

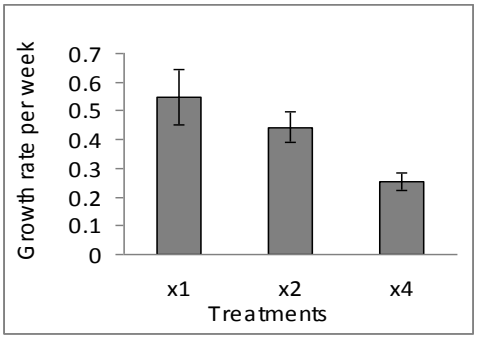


**Figure 10. Growth rate per week *A. inulifolium* under Low (x1), medium (x2) and high density (x4) treatments. Different letters indicate significant differences between treatments at p<0.05 levels; p=0.012, F=5.24, ANOVA.**

## Conclusions

*A. inulifolium* seems to be positively responding to unpredictable nutrient pulses and this character is important for their invasive nature. Fire incidents are common in the Knuckles Conservation Area, hence can result unpredictable nutrient pulses. Therefore, ecosystems with frequent fire incidents could be more vulnerable towards invasion by *Austroeupatorium.* Results also suggest that *Austroeupatorium* is physiologically more plastic than morphologically making them aggressive in competing for soil nutrients with other slow growing native plant species. However, intra-specific competition seems to affect the overall growth of *Austroeupatorium*.

## References

Bakker, J.P. and Berendse, F. (1999): Constraints in the restoration of ecological diversity in grassland and heathland communities. Trends in Ecology and Evolution, 14: 63-68.

Caldwell, M.M., Richards, J.H., Manwaring, J.H. and Eissenstat, D.M. (1987): Rapid shifts in phosphate acquisition show direct competition between neighboring plants. Nature, 327: 615-616.

Casper, B.B. and Cahill, J.F. Jr (1998): Population-level responses to nutrient heterogeneity and density by *Abutilon theophrasti* (Malvaceae): an experimental neighborhood approach. American Journal of Botany, 85: 1680-1687.

Casper, B.B., Cahill, J.F., Jackson, R.B. (2000): Plant competition in spatially heterogeneous environments. In: Ecological Consequences of Habitat Heterogeneity (Eds: M. Hutchings, E. John, A. Stewart), pp. 111-130. British Ecological Society Symposium Series.

Chapin, F.S. (1980): The mineral nutrition of wild plants. Annual Review of Ecology and Systematics. 11: 233-260.

Cleverly, J.R., Smith, S.D., Sala, A. and Devitt, D.A. (1997): Invasive capacity of *Tamarix amosissima* in a Mojave Desert floodplain: the role of drought. Oecologia, 111: 12-18.

Dale, M.P., Causton, D.R. (1992): The ecophysiology of *Veronica chamaedrys*, *V. montana* and *V. officinalis*. III. Effects of shading on the phenology of biomass allocations- a field experiment. Journal of Ecology, 80: 505-515.

D'Antonio, C.M. and Vitousek, P.M. (1992): Biological invasions by exotic grasses, the grass/fire cycle, and global change. Annual Review of Ecology and Systematics, 23: 63-87.

Drake, J. A., Mooney, H. A., di Castri, F., Groves, R. H., Kruger, F.J., Rejma´nek, M., Williamson, M. (Eds). (1989). Biological Invasions. A Global Perspective. John Wiley and Sons, Chichester, U.K

Dudley, S.A., and Schmitt, J. (1996): Testing the adaptive plasticity hypothesis: Density-dependent selection on manipulated stem length in *Impatiens capensis*. American Naturalist, 147: 445-465.

Grace, J.B. and Tilman, D. (1990): Perspectives on Plant Competition. Academic Press, New York.

Holl, K.D. (2002): Effect of Shrubs on Tree Seedling Establishment in an Abandoned Tropical Pasture. Journal of Ecology, 90: 179-187

Hutchings, M.J. and de Kroon, H. (1994): Foraging in plants: the role of morphological plasticity in resource acquisition. Advances in Ecological Research, 25, 159-238.

Levine, J.M., Vila M., D'Antonio, C.M., Dukes, J.S., Grigulis, K., and Lavorel, S. (2003): Mechanisms underlying the impacts of exotic plant invasions. Proceedings of the Royal Society of London B 270: 775-781.

Mack, M.C. and D'Antonio, C.M. (1998): Impacts of biological invasions on disturbance regimes. Trends in Ecology and Evolution, 13: 195-198

Massimino, D., Andre, M., Richaud, C., Daguenet, J., Massimino, J. and Vivoli, J. (1981) : The effect of a day at low irradiance of a maize crop. Physiologia Plantarum, 51: 150-155.

McDowell, S.C.L. (2002): Photosynthetic characteristics of invasive and non-invasive species of Rubus (Rosaceae). American Journal of Botany, 89: 1431-1438.

McFadyen, R. C. (2003): Chromolaena in Southeast Asia and the Pacific. Agriculture: New Directions for a New Nation — East Timor (Timor-Leste) ACIAR Proceedings No. 113.

Meekins, J.F., and McCarthy, B.C. (2000): Responses of the biennial forest herb *Alliaria petiolata* to variation in population density, nutrient addition and light availability. Journal of Ecology, 88: 447-463.

Milchunas, D.G., and Lauenroth, W.K. (1995): Inertia in plant community structure state changes after cessation of nutrient enrichment stress. Ecological Applications, 5: 452-458.

Mooney, H.A., and Drake, J.A. (1989): Biological Invasions: a SCOPE Program Overview. Biological Invasions, John Wiley and Sons, New York, pp. 491-508.

Parker, I.M., Rodriguez, J. and Loik, M.E. (2003): An evolutionary approach to understanding the biology of invasions: local adaptation and general-purpose genotypes in the weed Verbascum thapsus. Conservation Biology, 17: 59-72.

Reynolds, J.F., and D'Antonio, C. (1996) : The ecological significance of plasticity in root weight ratio in response to nitrogen: opinion. Plant and Soil, 185: 75-97.

Robinson, D., Hodge, A., Griffiths, B.S., Fitter, A.H. (1999): Plant root proliferation in nitrogen-rich patches confers competitive advantage. Proceedings of the Royal Society (London) Series B 266, 431–435.

Sa Xiao, Shu-Yan, Chen, Lu-Qiang, Zhao and Gang, Wang. (2006): Density Effects on Plant Height Growth and Inequality in Sunflower Populations. Journal of Integrative Plant Biology 48: 513-519.

Sexton, J.P., McKay, J.K. and Sala, A. (2002): Plasticity and genetic diversity may allow salt cedar to invade cold climates in North America. Ecological Applications, 12: 1652–1660.

Schmitt, J., and Wulff, R.D. (1993): Light spectral quality, phytochrome and plant competition. Trends Ecology and Evolution, 8: 47-50.

Skinner, K.E. (2004): Tree seedling regeneration in *Camelia sinensis* estates: assessing the potential for forest restoration across the forest-grassland boundary in Knuckles Forest Reserve, central Sri Lanka. http://www.abdn.ac.uk/asia-link/images/katSkinnerthesis .pdf.

Vitousek, P. (1986): Biological invasions and ecosystem properties: can species make a difference? In: Ecology of biological invasions of North America and Hawaii (Eds: H.A. Mooney, J.A. Drake), pp. 163–176. Springer-Verlag, New York,

Vitousek, P.M. and Walker, L.R. (1989): Biological invasion by *Myrica faya* in Hawaii: plant demography, nitrogen fixation, ecosystem effects. Ecological Monographs, 59: 247-265.

Wedin, D.A. and Tilman, D. (1996): Influence of nitrogen loading and species composition on the carbon balance of grasslands. Science, 274: 1720-1723.

Weiner, J. (1990): Asymmetric competition in plant populations. Trends in Ecology and Evolution, 5: 360-364.

White T.A., Campbell B.D. and Kemp P.D. (1997): Invasion of temperate grassland by a subtropical annual grass across an experimental matrix of water stress and disturbance. Journal of Vegetation Science, 8: 847-854.

Williams, D.G., Mack, R.N. and Black, R.A. (1995): Ecophysiology of introduced *Pennisetum setaceum* on Hawaii: the role of phenotypic plasticity. Ecology, 76: 1569-1580.